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Architecting MLOps in the Cloud: From Theory to Practice

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***Abstract*—ML operations (MLOps) refers to a set of practices and tools that automate and combine model development and model operation. MLOps can enable organizations to successfully deploy and manage their ML models in production. The MLOps landscape is increasingly expanding with technological choices, and most public cloud providers offer MLOps platforms with different degrees of maturity. However, this rapidly growing landscape makes designing and implementing an MLOps system in the cloud challenging as the practitioners need to make numerous architectural design decisions, select between different decision options, select between tools/services for realizing a particular decision option, and configure and assemble the chosen tools/services. In this tutorial, we will present guidelines for designing and implementing MLOps for ML-enabled applications in the cloud. We will cover each phase in the MLOps lifecycle. In addition to design guidance, tutorial participants will be able to get hands-on experience in creating an MLOps solution using the Google Cloud Platform (GCP).**

***Index Terms*—Machine Learning Operations, MLOps, Architecture Design, Cloud**

1. Summary

Organizations are struggling to create value from their investment in artificial intelligence (AI) and machine learning (ML) due to the challenges of the operationalization of ML/AI models. For example, according to Algorithmia [1], only 22% of organizations could successfully deploy and use a model in production. Consequently, ML operations (MLOps) have recently gained academia and industry's attention. MLOps aims to support the continuous delivery of high-quality ML models from the development environment into production and then monitoring and maintaining them [2]. On the other hand, organizations increasingly use the cloud as their primary IT infrastructure [3]. The cloud providers offer the MLOps platform consisting of services that can support the common activities in the MLOps, from model experimentation to model deployment to model governance. The MLOps landscape is also exploding with third-party tools that can be combined with cloud providers' offerings. However, the diverse and volatile MLOps technology space creates a dilemma for organizations who wants to implement MLOps: how to decide between tools/services and how to adopt, customize, and assemble them to realize an end-to-end MLOps lifecycle [4].

This tutorial aims to guide practitioners in navigating the MLOPs landscape when designing and implementing MLOps solutions in the cloud. In particular, we will share the consolidated knowledge gathered from our recent research studies on MLOps in the industry [5], [6], [7], [8], teaching MLOps to students, and the relevant literature from other researchers [9], [10], [11]. This tutorial will first introduce MLOps and its lifecycle model, from model experimentation to model governance. Then, we will present the various architectural and technological options that are available to choose from, especially when organizations adopt a cloud-first strategy for their MLOps journeys. We consider three major public clouds: Google Cloud Platform (GCP), Amazon Web Services (AWS), and Microsoft Azure. Finally, we will use a set of hands-on activities to gradually teach the participants how to architect and implement an MLOps solution in GCP.

2. Goals and Intended Audience

With this tutorial, participants will be able to learn about:

- The basics of MLOps and the architectural decisions that practitioners need to make when implementing MLOps for their ML-enabled cloud applications.

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TABLE 1: Tutorial Structure

Session	Duration
MLOps: Motivation, Definition, Lifeclye Model, and Architecture	45 mins
Break	15 mins
Design and Technological Options for MLOps in the Cloud	45 mins
Break	15 mins
Cloud Providers: MLOps Capabilities and Reference Architectures	45 mins
Break	15 mins
Hands-on Session on Implementing MLOps in GCP	45 mins
Wrap-up	15 mins

- Tools/services of the public cloud providers that can be used to implement their MLOps solutions.
- Get hands-on experience using GCP to implement each phase of the MLOps lifecycle.

3. Key Takeaways

The key takeaway messages of this tutorial are:

- Public cloud providers increasingly offer the tools/services for implementing MLOps for ML-enabled applications; the tutorial offers an overview of available architecture options.
- Selecting a particular cloud provider and implementing the different phases of the MLOps lifecycle using the tools/services provided by the selected cloud provider poses several challenges for practitioners; the tutorial allows one to compare the tool/service options.
- The knowledge about the requirements, reference architectures, and design and technology options for MLOps can guide practitioners to design and implement MLOps more systematically; the tutorial offers an overview of such available material for further practice.

4. Tutorial's Implementation

This half-day tutorial will first cover the basics of MLOps, including the requirements and lifecycle model of MLOps, and the common architectural elements in an MLOps environment. Next, we will discuss design and technology options that should be considered when creating and instantiating an MLOps architecture in the cloud. We will also present cloud provider-agnostic and provider-specific reference architectures for MLOps. Finally, we will work through an implementation of MLOps for an ML application using the GCP cloud. The tutorial participants will be able to get hands-on experience by deploying and executing this MLOps implementation. Table 1 shows the tutorial structure.

We will provide the participants with the presentation slides, the source code of the MLOPs implementation, and a manual and video for deploying and executing the implementation. Regarding getting access to the GCP cloud, we will provide a limited number of GCP credits. Google also offers free GCP credits for new users.

5. Presenters

Indika Kumara is an Assistant Professor at the Jheronimus Academy of Data Science (JADS) and Tilburg University, the Netherlands. He received his PhD from Swinburne University of Technology, Australia. His research interests include service-oriented computing, cloud computing, software architecture, and ML engineering. He teaches data engineering, MLOps, microservices, and data architecture at JADS's master program. In addition, he has been using the GCP cloud for the tutorials of his courses for the last six years.

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